

Building a Home Lab Operations System: HomeLab Edition

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Introduction

The cybersecurity landscape is becoming more dynamic and complex with each passing year. Organizations of all sizes face threats ranging from phishing and ransomware to sophisticated nation-state attacks. To keep pace, many enterprises rely on a Security Operations Center (SOC), which acts as the centralized hub for monitoring, detecting, and responding to cybersecurity incidents.



For a student or aspiring IT professional, recreating the structure and processes of a SOC inside a home lab provides an invaluable learning experience. A home lab SOC allows for hands-on practice with tools used in enterprise environments, develops the ability to understand how attacks unfold, and provides a safe environment to test defensive and response strategies. This paper will outline the goals, implementation process, and expected outcomes of building a SOC within a home lab setting.



Goals of the Home Lab SOC

The main objective of this project is to simulate real-world security operations in a controlled environment. Several goals stand out as central to the success of this build.

1. Develop Practical Skills

Theory alone is not enough to understand cybersecurity. Employers consistently look for candidates who have hands-on experience with the tools and workflows that SOC analysts and administrators use. By building a SOC lab, students gain practice with SIEM platforms such as Wazuh or Security Onion, endpoint monitoring tools like Sysmon and OSQuery, and intrusion detection systems such as Suricata and Zeek.

Personal skills	Social skills	Methodical skills
• Resilience • Responsibility • Engagement • Motivation • Inquisitiveness • Self-discipline • Self-reflection • Confidence	• Empathy • Integrability • Ability to communicate well • Ability to receive criticism • Knowledge of human nature • Team player • Interaction ability	• Analytical skills • Organizational talent • Presentation skills • Problem-solving skills • Stress resistance • Dealing with new media

2. Simulate Enterprise Environments

A SOC does not exist in isolation—it is part of an enterprise environment that includes endpoints, servers, firewalls, and users. The lab will replicate these components by deploying virtual machines that mimic both user workstations and attacker systems. For example, a Windows 11 client machine with Sysmon installed can represent a typical employee computer, while a Kali Linux VM can serve as the red team or attacker role. Simulating these scenarios allows the builder to experience both the defender's and attacker's perspectives.

3. Learn Threat Hunting Methodologies

One of the most valuable aspects of a SOC is the ability to go beyond reactive alerts and proactively search for signs of compromise. Threat hunting involves looking for anomalies in logs, traffic, and behavior that may indicate malicious activity. Using frameworks like MITRE ATT&CK, the lab will provide an opportunity to map observed activity to known attacker tactics and techniques, building both familiarity and confidence in the hunting process.

4. Practice Incident Response and Automation

Modern SOCs are not only about monitoring but also about responding quickly. This means creating structured workflows (runbooks) and, where possible, automating them to reduce response times. Open-source platforms such as TheHive and Cortex will serve as the foundation for incident response management, while Shuffle.io can be used to test automation playbooks. Even simple automated responses, such as blocking an IP address through a pfSense firewall rule, will showcase how orchestration can improve security operations.



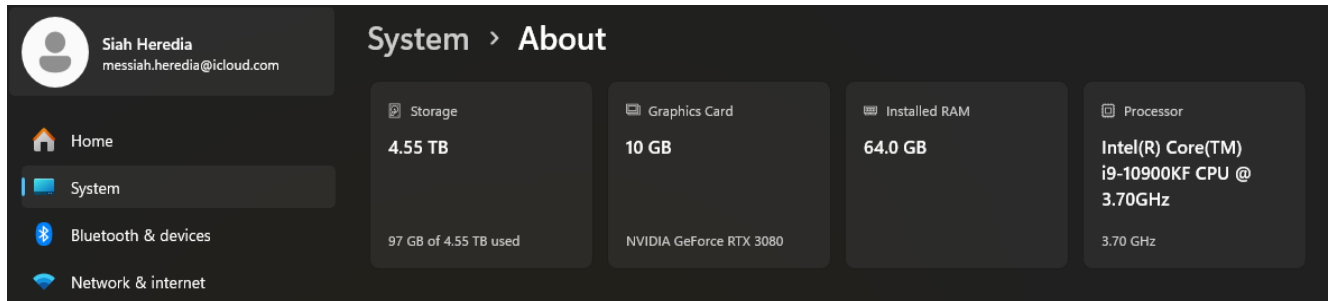
Implementation Process

The lab will be built in a series of steps that mirror enterprise SOC deployment. Each step contributes to the overall workflow of detection, analysis, and response.

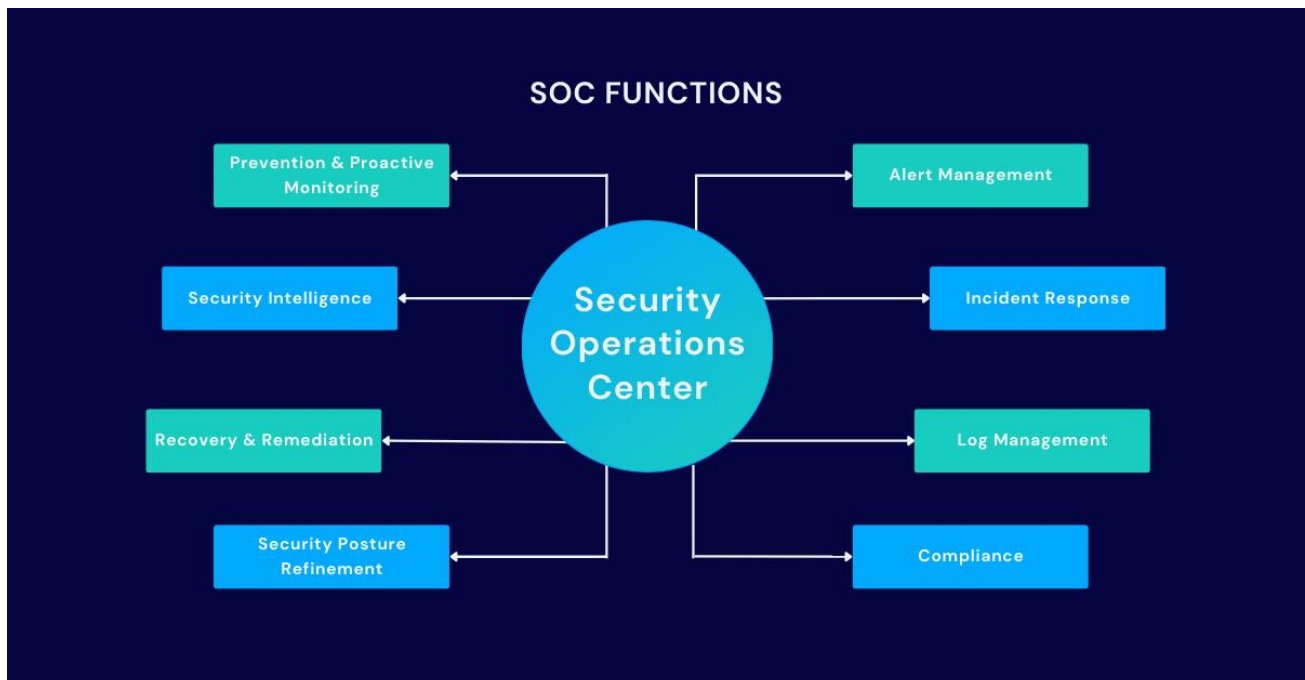
Infrastructure Setup

The foundation of the SOC will be a virtualized environment hosted on existing hardware. A high-performance rig with an Intel i9 processor and RTX 3080 GPU will serve as the primary virtualization host. Several virtual machines will be created:

- **VM1:** Ubuntu Server running Wazuh (SIEM).
- **VM2:** Security Onion (optional all-in-one monitoring).
- **VM3:** Windows 11 client machine with Sysmon installed.
- **VM4:** Kali Linux attacker machine for red team simulations.
- **VM5:** pfSense firewall VM for network monitoring and automated blocking.

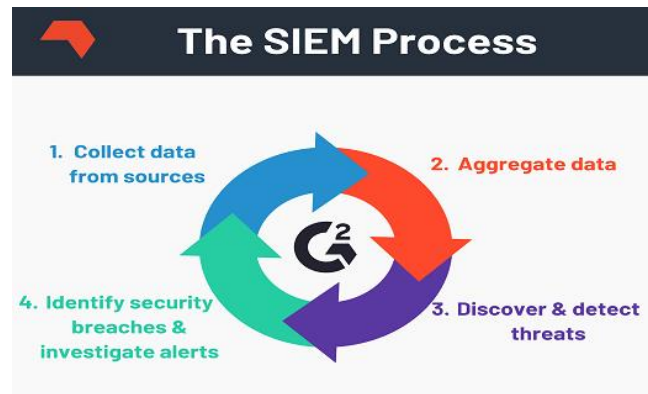


This setup creates a microcosm of a corporate network, allowing for realistic traffic flow, attacks, and defenses.



SIEM Deployment

The SIEM (Security Information and Event Management) platform will be the heart of the SOC. Wazuh, an open-source solution built on the Elastic Stack, will be installed on an Ubuntu VM. Wazuh is capable of aggregating logs from endpoints, servers, and network devices, and then correlating them for suspicious activity. Dashboards and alerting mechanisms will allow for real-time visibility into what is happening across the lab environment.



Endpoint Monitoring

To gain visibility into the activities of individual machines, endpoint monitoring tools will be deployed. On Windows systems, Microsoft Sysmon will capture detailed information such as process creation, file modifications, and network connections. On Linux systems, OSQuery will provide a SQL-like interface for querying processes, user activity, and system state. Both sets of logs will be forwarded into the SIEM for correlation and analysis.

Network Visibility

While endpoint monitoring is essential, attacks can often be identified by observing network traffic. Tools like Zeek and Suricata will be deployed to monitor network flows, inspect packets, and detect malicious patterns. Suricata, for instance, can identify traffic signatures related to malware, command-and-control channels, or known exploits. The combination of endpoint and network data will provide a full picture of any attack in progress.

Threat Simulation

To test the SOC, simulated attacks will be launched from the Kali Linux machine. These may include brute-force login attempts, phishing payloads, or even ransomware simulations. Tools such as Atomic Red Team will provide lightweight scripts that mimic real attacker behaviors as documented in the MITRE ATT&CK framework. The purpose of these exercises is not to cause damage but to generate realistic data that the SOC must detect, analyze, and respond to.

Incident Handling and Automation

When an attack is detected, the SOC workflow shifts to incident response. Alerts will be triaged using TheHive, which provides case management features for organizing investigations. Cortex

can be integrated to analyze suspicious files, hashes, or URLs. Shuffle.io will allow the creation of automated playbooks, such as automatically disabling a compromised account or isolating a machine from the network. Even if automation is limited in a lab setting, it demonstrates the principles of modern SOC orchestration.

Expected End Results

The completion of this lab SOC will produce several tangible outcomes.

Hands-On SOC Experience

The builder will gain practical exposure to the exact tools and workflows used in professional SOCs. This experience will be highly valuable when seeking internships or full-time roles, as it demonstrates initiative and technical competence beyond classroom theory.

Documented Attack-and-Defense Scenarios

Each simulated attack and the corresponding defense will generate artifacts such as log entries, SIEM alerts, and case reports. These can be documented as lab reports or visualized through screenshots of dashboards. Over time, this collection will form a portfolio of real-world security operations experience.

Proficiency in Security Tools

By the end of the build, the student will be comfortable deploying, configuring, and using key SOC tools: Wazuh, Security Onion, Sysmon, OSQuery, Zeek, Suricata, TheHive, and Shuffle.io. This skill set aligns closely with what employers look for in SOC analysts and system administrators.

Preparedness for Industry Roles

Most importantly, the lab will provide confidence and readiness for professional environments. Whether pursuing a role as a SOC analyst, security engineer, or systems administrator, the ability to explain and demonstrate a functioning SOC lab will stand out in interviews and professional conversations.

Conclusion

A home lab Security Operations Center is both a technical project and a professional investment. It simulates the workflows that enterprises depend on to keep their systems secure, while giving the builder a chance to learn by doing. By combining SIEM integration, endpoint and network

monitoring, threat hunting, and incident response automation, the lab will showcase the entire lifecycle of detecting and responding to threats.

The end result is a functioning SOC environment that not only deepens technical skills but also strengthens a professional portfolio. For a student or early-career IT professional, this lab represents a bridge between academic knowledge and real-world application, preparing them for the challenges of today's cybersecurity landscape.

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